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AUTOPLAYING CONTENT ACROSS MULTIPLE PROVIDERS ON A TELEVISION

Abstract

According to an aspect, a method may include receiving a request to generate an autoplay list, identifying a plurality of media content items, identifying a runtime for each of the plurality of media content items, based on determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime, generating a clip of the media content item with a runtime that is less than or equal to the predefined runtime, generating an ordered list of the plurality of media content items, substituting the clip for the media content item in place of the media content item included in the ordered list, generating an autoplay list based on the ordered list, and sending the autoplay list for a television application to play the plurality of media content items in the autoplay list on a network-connected display device.

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Background/Summary

BACKGROUND

[0001] A television (TV) application may present various types of media content of interest to a user. The media content may have different formats such as streaming video and audio. The types of media content may include, but are not limited to, movies, television shows, sporting events, news items, short form videos, and music. In addition, or in the alternative, a variety of media content providers may deliver various types of media content for viewing by the user. The TV application may deliver a customized viewing experience to a user that spans the diverse types of media content provided by the variety of media content providers.

SUMMARY

[0002] In some non-limiting examples, a user may autoplay media content on a television. The user may be presented with media content items that are selected and customized for the user based on preferences of the user and past viewing behaviors and habits of the user. The user may play and watch the media content without having to interact with a remote control or other device to select and advance the playing of the media content items. The user may autoplay a variety of types of media content that may include but are not limited to short form videos, movie clips, live television shows, live television news, live television sports, on demand video content, and television show episodes. Different media content providers may provide or stream the media content items to the television. In some implementations, the television may be a smart television that executes a television (TV) application. The TV application may interface with the media content providers to continuously stream recommended media content items to the television for autoplaying to the user.

[0003] In some aspects, the techniques described herein relate to a method including: receiving, by a server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0004] In some aspects, the techniques described herein relate to a method, wherein the network-connected display device is a smart television.

[0005] In some aspects, the techniques described herein relate to a method, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip.

[0006] In some aspects, the techniques described herein relate to a method, wherein substituting the clip for the media content item in place of the media content item includes serving the modified watch action in place of the media content item.

[0007] In some aspects, the techniques described herein relate to a method, wherein generating the clip of the media content item includes generating a first clip including: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.

[0008] In some aspects, the techniques described herein relate to a method, wherein generating the clip of the media content item further includes generating a second clip including: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.

[0009] In some aspects, the techniques described herein relate to a method, wherein substituting the clip for the media content item in place of the media content item includes substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.

[0010] In some aspects, the techniques described herein relate to a method, wherein identifying a plurality of media content items includes identifying a plurality of media content providers that provide media content to a user of the television application.

[0011] In some aspects, the techniques described herein relate to a method, wherein identifying a plurality of media content items includes determining preferences of a user for types of media content.

[0012] In some aspects, the techniques described herein relate to a method, wherein creating an ordered list of the plurality of media content items includes ordering the plurality of media content items in the ordered list based on the preferences of the user.

[0013] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium storing executable instructions that when executed by at least one processor of a server computer cause the at least one processor to execute operations, the operations including: receiving, by the server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0014] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium, wherein the network-connected display device is a smart television.

[0015] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip.

[0016] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium, wherein substituting the clip for the media content item in place of the media content item includes serving the modified watch action in place of the media content item.

[0017] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium, wherein generating the clip of the media content item includes generating a first clip including: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.

[0018] In some aspects, the techniques described herein relate to a non-transitory computer-

readable medium, wherein generating the clip of the media content item further includes generating a second clip including: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.

[0019] In some aspects, the techniques described herein relate to a non-transitory computer-readable medium, wherein substituting the clip for the media content item in place of the media content item includes substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.

[0020] In some aspects, the techniques described herein relate to a system including: at least one processor; and a non-transitory computer-readable medium storing instructions that when executed by the at least one processor cause the system to: receive a request to generate an autoplay list; and in response to receiving the request: identify a plurality of media content items; identify a runtime for each of the plurality of media content items; determine that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generate a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generate an ordered list of the plurality of media content items; substitute the clip for the media content item in place of the media content item included in the ordered list; generate the autoplay list based on the ordered list of the plurality of media content items; and send, to a network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0021] In some aspects, the techniques described herein relate to a system, wherein the network-connected display device is a smart television.

[0022] In some aspects, the techniques described herein relate to a system, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip; and wherein substituting the clip for the media content item in place of the media content item includes serving the modified watch action in place of the media content item.

[0023] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. 1A illustrates an example of a user interacting with a network-connected display device and a media adapter, according to implementations described throughout this disclosure.

[0025] FIG. 1B illustrates an example system for autoplating media content provided by different media content providers on a network-connected display device, according to implementations described throughout this disclosure.

[0026] FIG. 2 illustrates an example flow diagram of a process for autoplating different types of media content from different media content providers, according to implementations described throughout this disclosure.

[0027] FIG. 3 illustrates an example user interface for a television application autoplating media content on a television according to implementations described throughout this disclosure.

[0028] FIG. 4 illustrates a flowchart depicting example operations of generating an autoplay list for autoplating media content on a television according to implementations described throughout this

disclosure.

DETAILED DESCRIPTION

[0029] The disclosure generally relates to systems and methods for autoplating media content items on a television. In some implementations, a user may indicate that they would like a television application to autoplay media content of various different types and from a variety of different media content providers on a television. The user may want to watch a sequential and uninterrupted stream of media content items on the television. The user may prefer not to select or otherwise choose the media content items for autoplating, allowing the TV application to facilitate the selection of the media content of interest to the user. The TV application may provide recommended media content items for viewing by the user based on user preferences, and multi-dimensional user activity characteristics associated with an account of the user.

[0030] The user may play and watch the media content without having to interact with a remote control to select and advance the playing of the media content items. In some implementations, the user may interact with a remote control for a network-connected display device such as a smart television when selecting and watching media content items. In some implementations, a media adapter may provide media content to a network-connected display device. In some implementations, a mobile computing device of the user may implement a virtual remote control for interacting with the media adapter. In these implementations, the user may interact with the remote control device and/or the virtual remote control when selecting and viewing media content on the network-connected display device.

[0031] A user may autoplay a variety of types of media content provided by different media content providers on a network-connected display device. In some implementations, the network-connected display device may be a smart television that executes a television (TV) application. The TV application may interface with the media content providers to continuously stream recommended media content items to the television for autoplating to the user. In some implementations, a media adapter may interface with a server-side TV application. The server-side TV application may interface with the media content providers to provide recommended media content items to the media adapter for continuously streaming on the network-connected display device for autoplating to the user.

[0032] The TV application may be presented with the at least one technical problem of autoplating recommended media content items provided by multiple different media content providers. For example, different media content provider applications may provide the media content items for autoplating. As at least one technical solution to the at least one technical problem, the TV application may switch between media content provider applications when providing the media content items for autoplating. The TV application may switch between the media content provider applications without user intervention. The at least one technical effect may be the ability of the TV application to autoplay recommended media content items from different media content providers on a television without user interactions or interventions with the TV application.

[0033] FIG. 1A illustrates an example of a user **101** interacting with a network-connected display device **104** and a media adapter **107**, according to implementations described throughout this disclosure. FIG. 1B illustrates an example system **100** for generating an autoplay list of media content items from different media content providers for autoplating by a unified television application **130** executing on the network-connected display device **104**, according to implementations described throughout this disclosure.

[0034] The network-connected display device **104** may communicate with a server computer **106** and media content providers **160** by way of a network **150**. The media content providers **160**, the network-connected display device **104**, the server computer **106**, and a mobile computing device **102** may interact with and communicate with one other by way of the network **150**. In some implementations, the mobile computing device **102** may interface or connect to the media adapter **107** and/or the network-connected display device **104** by way of a wireless communication link that

may be a short-range wireless connection such as, for example a Bluetooth connection or a Wi-Fi (e.g., direct Wi-Fi) connection.

[0035] In some implementations, the user **101** may connect to and interact with the media adapter **107** by way of the network-connected display device **104** using a server-side television (TV) application **116** installed on the server computer **106**. The media adapter **107** may be connected or interfaced to the network-connected display device **104**. The network-connected display device **104** may be communicatively coupled or connected to the server computer **106** by way of the network **150**. In these implementations, a unified media platform (UMP) **158** may provide or serve media content items from the media content providers **160** to the network-connected display device **104** by way of the media adapter **107**.

[0036] In some implementations, the user **101** may interact with the network-connected display device **104** using a remote control device **105**. In some implementations, a television (TV) application **110** may render a virtual remote control **138** in a user interface (e.g., the UI **114**) on a display (e.g., the mobile computing device display **108**) on the mobile computing device **102**. The virtual remote control **138** may allow the mobile computing device **102** to act as a remote control for the network-connected display device **104**. The TV application **110** may render the virtual remote control **138** for use with the network-connected display device **104**. The user may interact with the remote control device **105** and/or the virtual remote control **138** when selecting media content for viewing on the network-connected display device **104**.

[0037] In some implementations, the user **101** may connect to and interact with the media adapter **107** using the television (TV) application **110** installed on the mobile computing device **102**. In some implementations, the user **101** may connect to and interact with the media adapter **107** using a media adapter remote control device **103**. In some implementations, the TV application **110** may render the virtual remote control **138** for use with the media adapter **107**. The virtual remote control **138** may allow the mobile computing device **102** to act as a remote control for the media adapter **107**. The user **101** may interact with the virtual remote control **138** and/or the media adapter remote control device **103** when interacting with the media adapter **107**.

[0038] The network-connected display device **104** may execute the unified television application **130**. The unified television application **130** may interface with a server-side television (TV) application **116**. The unified television application **130** may present an autoplay option **109** in a user interface (UI) **112** on a display **132** included in the network-connected display device **104**. For example, the user **101** may interact with the remote control device **105** to click on or select the autoplay option **109**. Once selected, the unified television application **130** executing on the network-connected display device **104** may request an autoplay list for the user **101** from an autoplay list generator **162** on the server computer **106**. The network-connected display device **104** may receive the autoplay list from the server computer **106**. The unified television application **130** may request the media content item associated with the entries in the autoplay list from the UMP **158**. The UMP **158** may interface with the media content providers **160** to obtain and serve the media content item to the unified television application **130** executing on the network-connected display device **104**. The unified television application **130** may present or play the media content item on the display **132**.

[0039] The UMP **158** may interface with the unified television application **130** executing on the network-connected display device **104** and the media content providers **160** to sequentially provide the media content items associated with the entries in the autoplay list in the order of the entries in the autoplay list to the network-connected display device **104** without any input or intervention from the user **101**. In addition, or in the alternative, the unified television application **130** may provide the media content items associated with the entries in the autoplay list to the network-connected display device **104** without any interruption or break between the playing of the media content items.

[0040] The server computer **106** may include a knowledge module **166**. The knowledge module

166 may include information associated with media content items provided by the media content providers **160**. In some implementations, the knowledge module **166** may generate media content recommendations for associating with an account of a user based, in part, on a multi-dimensional user activity characteristic associated with the account of the user and the information associated with media content items provided by the media content providers **160**. The user activity characteristic associated with the account of the user may be obtained from a plurality of information sources that may include, but are not limited to, a search engine, a mapping application, and an online retailer. The information sources may provide activity data related to activities of the account of the user by way of a respective software program or application.

[0041] The knowledge module **166** may help the autoplay list generator **162** determine media content that may be useful and of interest to the user. In some implementations, the unified television application **130** may interface with the knowledge module **166** to provide information and data related to the past activities of the user when interacting with the unified television application **130**, the viewing history of the user, and/or the popularity of media content items of a type, classification, category, group or genre.

[0042] The mobile computing device **102** may be configured to execute the TV application **110**. The mobile computing device **102** may include the mobile computing device display **108** configured to display the UI **114**. A user may interact with the UI **114** to set up, control, and interact with the TV application **110**. In some implementations, as described, the TV application **110** may display the virtual remote control **138** in the UI **114** allowing the user **101** to interact with and control the network-connected display device **104** and/or the media adapter **107**.

[0043] The mobile computing device **102** may be any type of computing device that includes one or more processors (processor(s) **140**), one or more memory devices (memory device(s) **142**), and an operating system **144**. The mobile computing device **102** may be a smartphone, a tablet, a wearable device, a laptop computer, or a desktop computer. In some implementations, the operating system **144** may be system software that manages computer hardware, software resources, and provides common services for computing programs.

[0044] In some implementations, the mobile computing device **102** may be a tablet, a smartphone, or a wearable. In these implementations, the operating system **144** may be referred to as a mobile operating system. The mobile operating system may be configured to execute on devices that, in general, include display devices that may be smaller in size than, for example, a display device included in a laptop computer or a desktop computer. In some implementations, the mobile computing device **102** may be a laptop computer. In these implementations, the operating system may be referred to as a laptop or desktop operating system. In these implementations, the operating system **144** may be an operating system designed for a display that is larger in size than that included in a tablet, a smartphone, or a wearable.

[0045] In some implementations, the media adapter **107** (e.g., a casting device, a media streaming device, a media streaming player, a set-top box) may be interfaced with or connected to the network-connected display device **104**. The media adapter **107** may interact with and communicate with the media content providers **160**, the server computer **106**, and the mobile computing device **102** when providing media content to the network-connected display device **104**. In some implementations, the media adapter **107** may be embedded in and/or an integrated part of the network-connected display device **104**.

[0046] The media content providers **160** may include a variety of streaming service and media content sources and service platforms. The media adapter **107** may facilitate providing (e.g., streaming) media content (e.g., streaming video such as movies, TV shows, etc.) from one or more streaming services included in the media content providers **160** to the network-connected display device **104**. For example, the media adapter **107** may directly connect to a connector on the network-connected display device **104** by way of connection **165**. The media adapter **107** may provide digital video and/or audio to the network-connected display device **104**. For example, the

media adapter **107** may connect to a high-definition multimedia interface (HDMI) connector included in the network-connected display device **104**. Examples of the media adapter **107** may include, but are not limited to, a set-top box, a television box, and a streaming media adapter. [0047] In some implementations, the mobile computing device **102** may connect to or interface with the media adapter **107** by way of a wireless communication link **163b**. Wireless communication links **163a-e** may be short-range wireless connections such as a Bluetooth connection. In some examples, wireless communication links **163a-e** may be a Wi-Fi (e.g., direct Wi-Fi) connection.

[0048] The media adapter **107** may be any type of computing device that includes one or more processors (processor(s) **170**), one or more memory devices (memory device(s) **172**), and an operating system **174**. In some implementations, the processor(s) **170** may include a system on a chip (SoC). The SoC may include a central processing unit (CPU), a graphic processing unit (GPU), one or more memory interfaces, and one or more input/output interfaces and devices. In some implementations, the operating system **174** may be system software that manages computer hardware, software resources, and provides common services for computing programs.

[0049] The network-connected display device **104** may include the unified television application **130**. The unified television application **130** may keep a record of the interactions of the user with the media content received from the server computer **106**. The network-connected display device **104** may send the record of the interactions to the server computer **106** for use in determining media content recommendations for the user.

[0050] In some implementations, the network-connected display device **104** may be configured to execute the unified television application **130**. For example, the network-connected display device **104** may be a smart television. For example, a smart television may be a network-connected television that may connect to media content providers (e.g., media content providers **160**) by way of a network (e.g., the network **150**). The media content providers may source media content to the smart television. In these implementations, a user may interact with the unified television application **130** to access media content from the media content providers **160**. The unified television application **130** may interface with the server computer **106**, and specifically with the server-side TV application **116**. The unified television application **130** may provide similar functionality to the user as that provided by an application executing on the media adapter **107**. For example, executing the unified television application **130** by the network-connected display device **104** allows the network-connected display device **104** to obtain a media content recommendation stream from the server computer **106**.

[0051] The network-connected display device **104** may be configured to connect to the network **150**. In some implementations, the network-connected display device **104** is a television (e.g., a smart television (TV)). The network-connected display device **104** may include one or more processors (processor(s) **156**), one or more memory devices (memory device(s) **152**), and an operating system (OS) **154**. The operating system **154** may execute (or assist with executing) the unified television application **130**.

[0052] In some implementations, the operating system **154** may be a browser application. A browser application is a web browser configured to access information on the Internet by way of a network (e.g., the network **150**). A browser application may launch one or more browser tabs in the context of one or more browser windows in the browser application. In some implementations, the operating system **154** is a Linux-based operating system configured to execute (or assist with executing) the unified television application **130**.

[0053] The system **100** may include one or more server computers (e.g., the server computer **106**) configured to interface with the mobile computing device **102**, the media adapter **107**, the media content providers **160**, and the network-connected display device **104** by way of the network **150**. In some implementations, the network **150** may establish a wireless communication link between the network-connected display device **104**, the mobile computing device **102**, the media adapter

107, the media content providers **160**, and the server computer **106**.

[0054] The server computer **106** may include the unified media platform (UMP) **158**. The UMP **158** may facilitate the providing of the media content items for autoplaying on the network-connected display device **104**. The UMP **158** may manage the providing of the media content items associated with entries in the autoplay list to the network-connected display device **104**. In some implementations, the UMP **158** may manage the providing of the media content items from the media content providers **160** to the mobile computing device **102**. The UMP **158** may facilitate the providing of the media content items with entries in the autoplay list to the network-connected display device **104** such that the network-connected display device **104** may autoplay the media content items on the network-connected display device **104**. The network-connected display device **104** may autoplay the media content items without user input or user intervention. In addition, or in the alternative, the network-connected display device **104** may autoplay the media content items in a continuous uninterrupted manner.

[0055] The UMP **158** may facilitate the providing of media content items to the network-connected display device **104** as described herein. In some implementations, the UMP **158** may facilitate the providing of the media content item for a respective entry in the autoplay list to the media adapter **107**. The media adapter **107** may stream the media content item from the media content providers **160** to the network-connected display device **104**. In response to receiving an indication of a selection of the autoplay option **109**, the server-side television application **116** may enable the sequential, uninterrupted display of the media content items associated with respective entries in the autoplay list on the display **132**. In some implementations, the UMP **158** may function as a centralized media content management module configured to facilitate providing media content items to the mobile computing device **102**.

[0056] The server computer **106** may include the server-side TV application **116**. The server-side TV application **116** may facilitate providing the media content items associated with the entries in the autoplay list to the unified television application **130** for playing on the network-connected display device **104**. The server-side TV application **116** may facilitate providing the media content items associated with the entries in the autoplay list to the unified television application **130** in response to receiving an indication from the unified television application **130** that a user selected the autoplay option **109**.

[0057] The server computer **106** may include a recommendation module **124**. The recommendation module **124** may interface with the knowledge module **166** and an artificial intelligence (AI) module **194** to determine media content items that may be recommended for autoplaying. The recommendation module **124** may provide the recommended media content items to the autoplay list generator **162** for inclusion in an autoplay list. In some implementations, the recommendation module **124** may provide a score or other type of indication that relates to an affinity of the user for the recommended media content item. The autoplay list generator **162** may sort the received recommended media content items based on the affinity of the user for the media content item, ranking the recommended media content items based on an associated score. The ranking may include the recommended media content items with a higher score or greater affinity of the user higher in the autoplay list.

[0058] The server computer **106** may include a trim module **148**. The trim module **148** may interface with the media content providers **160**, the UMP **158**, and the AI module **194**. In some implementations, the autoplay list generator **162** may specify a total runtime for a media content item included in the autoplay list. In some implementations, a media content item may have a runtime longer than the total runtime specified by the autoplay list generator **162**. The trim module **148** may parse the media content item to determine one or more clips that satisfy the specified runtime for a media content item in the autoplay list. The trim module **148** may provide the information about a clip to the autoplay list generator **162**. The autoplay list generator **162** may substitute the information about the clip for the media content item in place of the entry for the

media content item in the autoplay list. For example, the unified television application **130** may autoplay the clip for the media content item when autoplaying the media content items in the autoplay list.

[0059] In some implementations, the trim module **148** may provide the information about more than one clip (e.g., two or more clips) to the autoplay list generator **162**. The autoplay list generator **162** may substitute the information about the two or more clips for the media content item in place of the entry for the media content item in the autoplay list. For example, the unified television application **130** may autoplay the two or more clips for the media content item when autoplaying the media content items in the autoplay list.

[0060] The server computer **106** may include the artificial intelligence (AI) module **194**. The AI module **194** may receive information and data from the mobile computing device **102** and/or the network-connected display device **104** to build generative artificial intelligence (AI) model(s) **164** for use by the AI module **194**. In some implementations, the AI module **194** may provide the recommendation module **124** with a plurality of recommended media content items for use by the autoplay list generator **162**. The autoplay list generator **162** may include an entry for the recommended media content items in the autoplay list. The autoplay list generator **162** may provide the autoplay list to the server-side TV application **116**. The server-side TV application **116** may interface with the UMP **158** to provide the media content items associated with the entries in the autoplay list to the unified television application **130**.

[0061] The AI module **194** may receive updated media content recommendations from the knowledge module **166** along with updated information and data from the mobile computing device **102** and/or the network-connected display device **104** to retrain the generative AI model(s) **164**. The AI module **194** may use the retrained generative AI model(s) **164** to provide the autoplay list generator **162** with updated recommended media content items for inclusion in an autoplay list. The autoplay list generator **162** may generate an updated autoplay list for use by the server-side TV application **116**.

[0062] In some implementations, the generative AI engine **146** may use generative artificial intelligence along with the generative AI model(s) **164** for determining a start time and end time for a clip. For example, the generative AI model(s) **164** may determine one or more clips for a media content item that may be considered famous, liked, and/or enjoyed by users. In some implementations, the AI module **194** may be referred to as a generative artificial intelligence (generative AI) back end. For example, as described, generative AI may determine one or more clips in a media content item. The generative AI may curate or create a collection of clips for a media content item.

[0063] The use of generative AI can help further focus and refine the clips generated for a media content item. For example, the generative AI engine **146** may update and refine the generative AI model(s) as more knowledge is gained about the clips by receiving responses and feedback from users who view the clips.

[0064] The mobile computing device **102** may include the mobile computing device display **108**. In some implementations, the mobile computing device display **108** is a display device such as a liquid crystal display (LCD), a light-emitting diode (LED) display, an organic light-emitting diode (OLED) display, or an active-matrix organic light-emitting diode (AMOLED) display. The network-connected display device **104** may include the display **132**. In some implementations, the display **132** is a display device such as a liquid crystal display (LCD), a light-emitting diode display (LED) display, a plasma display, a quantum dot light-emitting diode display (QLED) display, or an organic light-emitting diode (OLED) display.

[0065] The processor(s) **156**, the processor(s) **140**, the processor(s) **170**, and the processor(s) **180** may be formed in a substrate configured to execute one or more machine executable instructions or pieces of software, firmware, or a combination thereof. The processor(s) **156**, the processor(s) **140**, the processor(s) **170**, and the processor(s) **180** may be semiconductor-based. For example, the

processor(s) **156**, the processor(s) **140**, the processor(s) **170**, and the processor(s) **180** may include semiconductor material that can perform digital logic.

[0066] The memory device(s) **152**, the memory device(s) **142**, the memory device(s) **172**, and the memory device(s) **182** may include main memory that stores information in a format that can be read and/or executed by the processor(s) **156**, the processor(s) **140**, the processor(s) **170**, and the processor(s) **180** respectively. The memory device(s) **152**, the memory device(s) **142**, the memory device(s) **172**, and the memory device(s) **182** may include one or more random-access memory (RAM) devices and/or one or more read-only memory (ROM) devices.

[0067] The memory device(s) **152**, memory device(s) **142**, the memory device(s) **172**, and the memory device(s) **182** may store applications that, when executed by the processor(s) **156**, the processor(s) **140**, the processor(s) **170**, and the processor(s) **180**, respectively, perform operations. For example, the memory device(s) **142** may store the operating system **144** and the TV application **110** that, when executed by the processor(s) **140**, may perform operations on the mobile computing device **102**. For example, the memory device(s) **152** may store the operating system **154** and the unified television application **130** that, when executed by the processor(s) **156**, may perform operations on the network-connected display device **104**.

[0068] In some implementations, the memory device(s) **182** may represent any kind of (or multiple kinds of) memory (e.g., RAM, flash, cache, disk, tape, etc.). In some implementations, the memory device(s) **182** may include external storage, e.g., memory physically remote from but accessible by the server computer **106**. The server computer **106** may include one or more modules, engines, or applications representing specially programmed software. In some implementations, the server computer **106** may include the operating system **184**, the server-side TV application **116**, the knowledge module **166**, the AI module **194**, the generative AI engine **146**, the generative AI model(s) **164**, the UMP **158**, the autoplay list generator **162**, the trim module **148**, the recommendation module **124**, processor(s) **180**, and the device(s) **182**. For example, the memory device(s) **182** may store the operating system **184**, the server-side TV application **116**, the knowledge module **166**, the AI module **194**, the generative AI engine **146**, the generative AI model(s) **164**, the UMP **158**, the autoplay list generator **162**, the trim module **148**, the recommendation module **124** that, when executed by the processor(s) **180**, may perform operations on server computer **106** to implement one or more of the methods and processes described herein.

[0069] The network **150** may include the Internet and/or other types of data networks, such as a local area network (LAN), a wide area network (WAN), a cellular network, satellite network, or other types of data networks. The network **150** may also include any number of computing devices (e.g., computer, servers, routers, network switches, etc.) that are configured to receive and/or transmit data within the network **150**. The network **150** may further include any number of hardwired and/or wireless connections. The network **150** may be, for example, communications networks having one or more types of topologies, including but not limited to the Internet, intranets, local area networks (LANs), cellular networks, Ethernet, Storage Area Networks (SANs), telephone networks, and Bluetooth personal area networks (PAN). In some implementations, two or more devices in a sub-network may be coupled by way of a wired connection, while at least some of the devices in the same sub-network are coupled by way of a local radio communication network (e.g., ZigBee, Z-Wave, Insteon, Bluetooth, Wi-Fi and other radio communication networks).

[0070] FIG. 2 illustrates an example flow diagram of a process **200** for autoplating different types of media content **202a-f** from different media content providers **160**, according to implementations described throughout this disclosure. Referring with reference to FIGS. 1A and 1B, the unified television application **130** may provide a user interface **118** for display in the UI **112** on the display **132**. The user **101** may select or click on the autoplay option **109** that may begin autoplating recommended media content items included in an autoplay list on the network-connected display device **104**. In response to receiving an indication of the selection of the selection of the autoplay option **109**, the unified television application **130** may create a request for an autoplay list for the

user **101**. The network-connected display device **104** may send the request for the autoplay list to the server computer **106**. The server-side TV application **116** may receive the request for the autoplay list and may interface with the autoplay list generator **162** that may create or generate the autoplay list for sending by the server computer **106** to the network-connected display device **104** for use by the unified television application **130** when serving the watch list of media content items included in the autoplay list to the display **132** for viewing by the user **101** without user intervention or input.

[0071] The media content providers may provide a variety of different types of media content that may be of interest to a user. For example, the media content providers **160** may provide media content that may include but is not limited to movies **202a**, television show episode(s) **202b**, video sharing content **202c**, live news content **202d**, live television content **202e**, and live sports content **202f**.

[0072] The knowledge module **166** may receive information and data related to the media content **202a-f**. The information and data may be metadata associated with a media content item. The metadata may include but is not limited to a runtime for the media content item, actors, titles, descriptions, content providers, genres, etc. The recommendation module **124** may receive media content recommendations associated with an account of a user from the knowledge module **166**. The recommendation module **124** may determine the media content items to recommend based, in part, on a multi-dimensional user activity characteristic associated with the account of the user and the information and data associated with media content items provided by the media content providers **160**. The user activity characteristic associated with the account of the user may be obtained from a plurality of information sources that may include, but are not limited to, a search engine, a mapping application, and an online retailer. The information sources may provide activity data related to activities of the account of the user by way of a respective software program or application.

[0073] The recommendation module **124** may provide a list of recommended media content items for viewing by the user **101** to the UMP **158**. The recommendation module **124** may provide the metadata associated with a media content item to the recommendation module **124**. In some implementations, the recommendation module **124** may generate an ordered and ranked list of the recommended media content items with the more highly recommended media content items at the top of the list. The UMP **158** may manage the providing of the media content items associated with the media content recommendations to the network-connected display device **104**.

[0074] The recommendation module **124** may provide the metadata associated with each respective media content item included in the list of the recommended media content items to the UMP **158**. In some implementations, the UMP **158** may receive a runtime for a media content item for inclusion in an autoplay list from the autoplay list generator **162**. For example, the autoplay list generator **162** may determine a preferred runtime for media content items for autoplating to a user. The preferred runtime may be based on user feedback related to viewing times for individual media content items autoplaid to a user. In some implementations, the preferred runtime may be the same for all types of media content, such as shared videos, movies, etc. In some implementations, the preferred runtime may vary depending on the type of media content. For example, a user may prefer a runtime for autoplating shared video content items that is less than a preferred runtime for autoplating a movie, which typically may have a longer total runtime as compared to a total runtime for a shared video. In some implementations, the preferred runtime for autoplating movie content may be less than the total runtime for the movie.

[0075] In some implementations, the UMP **158** may provide a recommended media content item to the generative AI model(s) **164**. In cases where the runtime of the recommended media content item exceeds a preferred runtime for the type of media content item, the generative AI model(s) **164** may determine one or more scenes included in the media content item that may be of interest to the user. The generative AI model(s) **164** may determine a start time and an end time in the

media content item that defines a scene in the media content item. The trim module **148** may use the start time and the end time for the scene to generate a clip for the media content item. The clip may define a modified watch action for the media content item that includes the start time and the end time for the scene. The clip may be substituted for the media content item in the autoplay list generated by the autoplay list generator **162**. In some implementations, the generative AI model(s) **164** may determine a start time and an end time for a plurality of scenes in the media content item. The trim module **148** may use the start time and the end time for each scene to generate respective clips for the plurality of scenes in the media content item. A clip may define a modified watch action for the media content item that includes the start time and the end time for the respective scene. In these implementations, the clips (e.g., two or more clips) may be substituted for the media content item in the autoplay list generated by the autoplay list generator **162**.

[0076] The autoplay list generator **162** may provide the autoplay list to the server-side TV application **116**. The server-side TV application **116** may interface with the media content providers **160** to serve, send, or provide the different types of media content **202a-f** from the different media content providers **160** included in the autoplay list to the network-connected display device **104** for autoplating to the user **101**.

[0077] FIG. **3** illustrates an example user interface **300** for a television application (e.g., the unified television application **130**) autoplating media content on a television (e.g., the network-connected display device **104**) according to implementations described throughout this disclosure. The user interface **300** includes a watch area **302** for autoplating the media content item. The user interface **300** may include an identification area **304** that may include metadata associated with the media content item playing in the watch area **302** such as a title.

[0078] The user interface **300** may include a next button **306**. For example, a user may want to skip the playing of the current media content item in the watch area **302** to advance to the playing of the next media content item in the watch area **302**. The user may select or click on the next button **306** to end the playing of the current media content item in the watch area **302** and advance to playing the media content item that is next in the autoplay list in the watch area **302**.

[0079] The user interface **300** may include an end button **308**. For example, the user may want to stop or end the autoplating of media content. The user may select or click on the end button **308** to stop autoplating media content in the watch area **302**.

[0080] FIG. **4** illustrates a flowchart depicting example operations of generating an autoplay list for autoplating media content on a television according to implementations described throughout this disclosure. Although the flowchart **400** of FIG. **4** illustrates the operations in sequential order, it will be appreciated that this is merely an example, and that additional or alternative operations may be included. Further, operations of FIG. **4** and related operations may be executed in a different order than that shown, or in a parallel or overlapping fashion. The operations may define a computer-implemented method. Although the flowchart **400** is described with reference to the system **100** of FIG. **1B**, the flowchart **400** may be executed according to any of the figures discussed herein. In some examples, the operations of the flowchart **400** are executed by server computer **106**.

[0081] Operation **402** includes receiving, by a server computer and from a network-connected display device, a request to generate an autoplay list. For example, a user may select or click on the autoplay option **109**. In response to the selection of the autoplay option **109**, the unified television application **130** may generate a request for the generation of an autoplay list that the network-connected display device **104** may send or transmit to the server computer **106**.

[0082] Operation **404** includes, in response to receiving the request, identifying a plurality of media content items. For example, the server-side TV application **116** may interface with the recommendation module **124**. The recommendation module **124** may identify the plurality of media content items as described herein.

[0083] Operation **406** includes identifying a runtime for each of the plurality of media content

items. For example, the UMP **158** may identify a runtime for each of the plurality of media content items.

[0084] Operation **408** includes determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime. For example, the UMP **158** may determine that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime.

[0085] Operation **410** includes, based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime. For example, the generative AI model(s) **164** may determine one or more scenes included in the media content item that may be of interest to the user. The trim module **148** may use the start time and the end time for the scene to generate a clip for the media content item.

[0086] Operation **412** includes generating an ordered list of the plurality of media content items. For example, the recommendation module **124** may generate an ordered list of the plurality of media content items.

[0087] Operation **414** includes substituting the clip for the media content item in place of the media content item included in the ordered list. For example, the clip may define a modified watch action for the media content item that includes the start time and the end time for the scene. The clip may be substituted for the media content item in the autoplay list generated by the autoplay list generator **162**.

[0088] Operation **416** includes generating the autoplay list based on the ordered list of the plurality of media content items. For example, the autoplay list generator **162** may generate the autoplay list based on the ordered list of the plurality of media content items

[0089] Operation **418** includes sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device. For example, the server computer **106** may send the autoplay list to the network-connected display device **104**. The unified television application **130** may receive the autoplay list and begin playing the media content items in the autoplay list on the display **132**.

[0090] In some examples, the techniques described herein relate to a method including: receiving, by a server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0091] In some examples, the techniques described herein relate to a method, wherein the network-connected display device is a smart television.

[0092] In some examples, the techniques described herein relate to a method, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip.

[0093] In some examples, the techniques described herein relate to a method, wherein substituting the clip for the media content item in place of the media content item includes serving the modified

watch action in place of the media content item.

[0094] In some examples, the techniques described herein relate to a method, wherein generating the clip of the media content item includes generating a first clip including: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.

[0095] In some examples, the techniques described herein relate to a method, wherein generating the clip of the media content item further includes generating a second clip including: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.

[0096] In some examples, the techniques described herein relate to a method, wherein substituting the clip for the media content item in place of the media content item includes substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.

[0097] In some examples, the techniques described herein relate to a method, wherein identifying a plurality of media content items includes identifying a plurality of media content providers that provide media content to a user of the television application.

[0098] In some examples, the techniques described herein relate to a method, wherein identifying a plurality of media content items includes determining preferences of a user for types of media content.

[0099] In some examples, the techniques described herein relate to a method, wherein creating an ordered list of the plurality of media content items includes ordering the plurality of media content items in the ordered list based on the preferences of the user.

[0100] In some examples, the techniques described herein relate to a non-transitory computer-readable medium storing executable instructions that when executed by at least one processor of a server computer cause the at least one processor to execute operations, the operations including: receiving, by the server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0101] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein the network-connected display device is a smart television.

[0102] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip.

[0103] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein substituting the clip for the media content item in place of the media content item includes serving the modified watch action in place of the media content item.

[0104] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein generating the clip of the media content item includes generating a first clip including: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.

[0105] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein generating the clip of the media content item further includes generating a second clip including: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.

[0106] In some examples, the techniques described herein relate to a non-transitory computer-readable medium, wherein substituting the clip for the media content item in place of the media content item includes substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.

[0107] In some examples, the techniques described herein relate to a system including: at least one processor; and a non-transitory computer-readable medium storing instructions that when executed by the at least one processor cause the system to: receive a request to generate an autoplay list; and in response to receiving the request: identify a plurality of media content items; identify a runtime for each of the plurality of media content items; determine that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generate a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generate an ordered list of the plurality of media content items; substitute the clip for the media content item in place of the media content item included in the ordered list; generate the autoplay list based on the ordered list of the plurality of media content items; and send, to a network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

[0108] In some examples, the techniques described herein relate to a system, wherein the network-connected display device is a smart television.

[0109] In some examples, the techniques described herein relate to a system, wherein generating the clip of the media content item includes creating a modified watch action for the media content item that includes a start time and an end time for the clip; and wherein substituting the clip for the media content item in place of the media content item includes serving the modified watch action in place of the media content item.

[0110] Various implementations of the systems and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0111] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” “computer-readable medium” refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or

data to a programmable processor, including a non-transitory machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. [0112] To provide for interaction with a user, the systems and techniques described here can be implemented on a computer having a display device (e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor) for displaying information to the user and a keyboard and a pointing device (e.g., a mouse or a trackball) by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback (e.g., visual feedback, auditory feedback, or tactile feedback); and input from the user can be received in any form, including acoustic, speech, or tactile input.

[0113] The systems and techniques described here can be implemented in a computing system that includes a back end component (e.g., as a data server), or that includes a middleware component (e.g., an application server), or that includes a front end component (e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the systems and techniques described here), or any combination of such back end, middleware, or front end components. The components of the system can be interconnected by any form or non-transitory medium of digital data communication (e.g., a communication network). Examples of communication networks include a local area network (“LAN”), a wide area network (“WAN”), and the Internet.

[0114] The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

[0115] In this specification and the appended claims, the singular forms “a,” “an” and “the” do not exclude the plural reference unless the context clearly dictates otherwise. Further, conjunctions such as “and,” “or,” and “and/or” are inclusive unless the context clearly dictates otherwise. For example, “A and/or B” includes A alone, B alone, and A with B. Further, connecting lines or connectors shown in the various figures presented are intended to represent example functional relationships and/or physical or logical couplings between the various elements. Many alternative or additional functional relationships, physical connections or logical connections may be present in a practical device. Moreover, no item or component is essential to the practice of the embodiments disclosed herein unless the element is specifically described as “essential” or “critical”.

[0116] Terms such as, but not limited to, approximately, substantially, generally, etc. are used herein to indicate that a precise value or range thereof is not required and need not be specified. As used herein, the terms discussed above will have ready and instant meaning to one of ordinary skill in the art.

[0117] Moreover, use of terms such as up, down, top, bottom, side, end, front, back, etc. herein are used with reference to a currently considered or illustrated orientation. If they are considered with respect to another orientation, it should be understood that such terms must be correspondingly modified.

[0118] Further, in this specification and the appended claims, the singular forms “a,” “an” and “the” do not exclude the plural reference unless the context clearly dictates otherwise. Moreover, conjunctions such as “and,” “or,” and “and/or” are inclusive unless the context clearly dictates otherwise. For example, “A and/or B” includes A alone, B alone, and A with B.

[0119] Although certain example methods, apparatuses and articles of manufacture have been described herein, the scope of coverage of this patent is not limited thereto. It is to be understood that terminology employed herein is for the purpose of describing particular aspects and is not intended to be limiting. On the contrary, this patent covers all methods, apparatus and articles of

manufacture fairly falling within the scope of the claims of this patent.

[0120] Further to the descriptions above, a user may be provided with controls allowing the user to make an election as to both if and when systems, programs, or features described herein may enable collection of user information (e.g., a user's preferences, a user's current location, a user's credentials, etc.), and if the user is sent content or communications from a server. In addition, certain data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over what information is collected about the user, how that information is used, and what information is provided to the user.

Claims

1. A method comprising: receiving, by a server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.
2. The method of claim 1, wherein the network-connected display device is a smart television.
3. The method of claim 1, wherein generating the clip of the media content item comprises creating a modified watch action for the media content item that includes a start time and an end time for the clip.
4. The method of claim 3, wherein substituting the clip for the media content item in place of the media content item comprises serving the modified watch action in place of the media content item.
5. The method of claim 1, wherein generating the clip of the media content item comprises generating a first clip comprising: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.
6. The method of claim 5, wherein generating the clip of the media content item further comprises generating a second clip comprising: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.
7. The method of claim 6, wherein substituting the clip for the media content item in place of the media content item comprises substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.
8. The method of claim 1, wherein identifying a plurality of media content items comprises identifying a plurality of media content providers that provide media content to a user of the television application.
9. The method of claim 1, wherein identifying a plurality of media content items comprises

determining preferences of a user for types of media content.

10. The method of claim 9, wherein creating an ordered list of the plurality of media content items comprises ordering the plurality of media content items in the ordered list based on the preferences of the user.

11. A non-transitory computer-readable medium storing executable instructions that when executed by at least one processor of a server computer cause the at least one processor to execute operations, the operations comprising: receiving, by the server computer and from a network-connected display device, a request to generate an autoplay list; and in response to receiving the request: identifying a plurality of media content items; identifying a runtime for each of the plurality of media content items; determining that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generating a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generating an ordered list of the plurality of media content items; substituting the clip for the media content item in place of the media content item included in the ordered list; generating the autoplay list based on the ordered list of the plurality of media content items; and sending, by the server computer and to the network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

12. The non-transitory computer-readable medium of claim 11, wherein the network-connected display device is a smart television.

13. The non-transitory computer-readable medium of claim 11, wherein generating the clip of the media content item comprises creating a modified watch action for the media content item that includes a start time and an end time for the clip.

14. The non-transitory computer-readable medium of claim 13, wherein substituting the clip for the media content item in place of the media content item comprises serving the modified watch action in place of the media content item.

15. The non-transitory computer-readable medium of claim 11, wherein generating the clip of the media content item comprises generating a first clip comprising: identifying a start time and an end time for a first scene included in the media content item; and creating a first modified watch action for the media content item that includes the start time and the end time for the first scene.

16. The non-transitory computer-readable medium of claim 15, wherein generating the clip of the media content item further comprises generating a second clip comprising: identifying a start time and an end time for a second scene included in the media content item; and creating a second modified watch action for the media content item that includes the start time and the end time for the second scene.

17. The non-transitory computer-readable medium of claim 16, wherein substituting the clip for the media content item in place of the media content item comprises substituting the first clip for the media content item and the second clip for the media content item in place of the media content item by serving the first modified watch action and the second modified watch action in place of the media content item.

18. A system comprising: at least one processor; and a non-transitory computer-readable medium storing instructions that when executed by the at least one processor cause the system to: receive a request to generate an autoplay list; and in response to receiving the request: identify a plurality of media content items; identify a runtime for each of the plurality of media content items; determine that a runtime for a media content item included in the plurality of media content items exceeds a predefined runtime; based on determining that the runtime for the media content item exceeds the predefined runtime, generate a clip of the media content item, the runtime of the clip being less than or equal to the predefined runtime; generate an ordered list of the plurality of media content items; substitute the clip for the media content item in place of the media content item included in

the ordered list; generate the autoplay list based on the ordered list of the plurality of media content items; and send, to a network-connected display device, the autoplay list for a television application, executing on the network-connected display device, to play the plurality of media content items in the autoplay list on the network-connected display device.

19. The system of claim 18, wherein the network-connected display device is a smart television.

20. The system of claim 18, wherein generating the clip of the media content item comprises creating a modified watch action for the media content item that includes a start time and an end time for the clip; and wherein substituting the clip for the media content item in place of the media content item comprises serving the modified watch action in place of the media content item.
